

Dynamic Multicasting in Optical Split-Incapable WDM Networks for Supporting High-Bandwidth Applications

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Abstract—With the advent of bandwidth intensive applications, the demand for multicast networking capabilities has become an essential component of wavelength division multiplexed (WDM) optical networks. Traditionally, the multicast functionality is accomplished by splitting a signal all-optically, thereby creating a *light-tree*, which originates from the source node and reaches a subset of the destination nodes. To support the multicasting functionality in an optical network that is *Split-Incapable*, i.e., the optical cross connects are incapable of switching an incoming optical signal to more than one output interface, one must implement a logical overlay to the underlying optical layer. A naive approach to accomplish this is by creating a set of unicast lightpaths that originate at the source node and terminate at a subset of the destination nodes of the multicast request. However, for a large number of requests this approach leads to a poor utilization of network resources. To overcome this problem, we propose two overlay approaches: Multicasting with Drop at Member Node (MA-DAMN) and Multicasting with Drop at Any Node (MA-DAAN). In these solutions, we achieve multicasting by creating a set of lightpath routes (possibly multiple hops) in the overlay layer. We consider a dynamic traffic model and propose efficient heuristics to solve the MA-DAMN and MA-DAAN problems with a goal of minimizing the total number of wavelengths required to satisfy the requests. Moreover, we present a simple heuristic to approximate the unicast approach of the naive method. Our results demonstrate that both the overlay approaches reduce the wavelength consumption by approximately 33 – 45% over the naive multicasting via WDM unicast approach for real-world large-scale networks.¹

Index Terms—Multicasting, WDM, Overlay, Lightpath Routes, Split-Incapable, O-E-O.

I. INTRODUCTION

Optical wavelength division multiplexed (WDM) networks have proven to be an important technology to provide high-bandwidth and services to the next-generation Internet. Multicasting has caught the attention of several researchers during the recent past, due to the emergence of many distributed applications [1], [2]. Distributed applications, such as video conferencing, distributed interactive simulations (DIS), grid computing, storage area network (SAN), and distributed content distribution network (CDN) applications require large amounts of bandwidth and an effective communication between a single source and a set of destinations. For example, the data generated by the Large Hadron Collider is frequently sent to multiple Department of Energy (DOE) laboratories across the United States over the DOE's Energy Sciences network (ESnet)². This data may be stored at multiple research

laboratories for distributed backup.

Multicasting [3], is a powerful communication framework that can be used to support such applications. Multicasting is a generic version of *Anycasting*³ that supports communication from a sender to any k out of m ($k \leq m$) candidate destinations where the candidate destination set, $|D_c| = m$, is a subset of nodes in the network. By changing the parameters of a multicast request, one can also perform unicast ($k = m = 1$), multicast ($k = m > 1$), and anycast ($k = 1 < m$). The key difference between multicast and multicast is that in multicast, all destinations are specified *a priori*, whereas in multicast only a subset of candidate destinations must be chosen. We can use multicasting to choose some subset of these locations, while ensuring that the underlying network is used efficiently.

Traditionally, one can support multicasting at the optical layer using a technique called *Light-Trees* [4], which employs light-splitters in the optical crossconnects (OXC). A fundamental problem in supporting multicast over networks like the ESnet is that the underlying optical-layer [5] is *Split-Incapable* i.e., the OXCs are incapable of switching an incoming optical signal to more than one output interface. To overcome this problem, one can implement a simple unicast approach, wherein we establish an independent point-to-point communication channel (lightpath) between the source and each selected candidate destination of the multicast request set. This naive *Multicasting via WDM Unicast (MA-VWU)* approach to the multicast overlay problem results in an inefficient consumption of bandwidth. As an alternative, we present two multi-hop solutions which will provide a set of lightpath-routes from source to every selected multicast destination using Steiner tree [6] routing: 1) *Multicasting with Drop At Member Node (MA-DAMN)*, which limits lightpath termination to only members (source or destination nodes) of the multicast request, and 2) *Multicasting with Drop At Any Node (MA-DAAN)*, which relaxes this constraint and allows a lightpath to be potentially terminated at any node in the network.

In this paper, we consider a dynamic traffic model, wherein the multicast requests arrive to the network according to some stochastic process⁴. We present heuristics to solve the multicast overlay problems for the MA-DAMN and MA-DAAN schemes, with the goal of minimizing the number of wavelengths required to satisfy all the multicast requests. We

¹This work was supported by the Department of Energy (DOE) COMMON project under grant DE-SC0004909.

²www.es.net : Energy Sciences Networks of the Department of Energy (DOE).

³Anycasting is defined as the communication paradigm in which the user has the ability to choose a single destination from a group of candidate destinations, unlike deciding it *a priori* as in unicast.

⁴We consider the static version of the multicast overlay problem in [7].

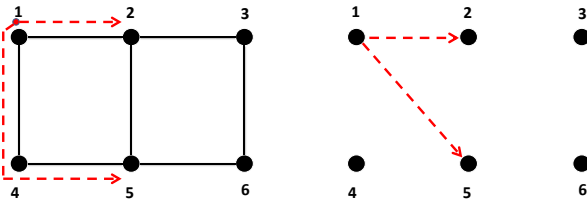
compare the performance of MA-DAMN and MA-DAAN with a simple heuristic to solve the MA-VWU problem. The remainder of this paper is structured as follows: In Section 2 we define the MA-VWU, MA-DAMN, and MA-DAAN problems formally and present the corresponding heuristics in Section 3. Simulation results are presented in Section 4, and Section 5 concludes the paper.

II. PROBLEM DEFINITIONS

Given a network topology graph $G(V, E)$, with V nodes and E links, a multicast request can be defined as $R = (s_r, D_r, K')$, where s_r is the source node of the request ($s_r \in V$) and D_r is the set of candidate destination members (nodes) ($D_r \subseteq V - \{s_r\}$). For a multicast request with K destination members, we represent the destination set as $D_r = \{d_1, d_2, \dots, d_K\}$. $K' \leq K$ is the number of nodes necessary to reach in order to successfully service the multicast request. In the MA-VWU problem, we establish lightpaths from the source node of the multicast request to each of the K' selected destination nodes. In doing so, we create K' unicast routes (single logical-hop) in the overlay network. In the MA-DAMN model, we find a set of lightpath routes that start at the source node and reach the K' destination members possibly via multiple logical-hops. While creating these lightpath routes, we take into account that we can terminate (i.e., drop) a lightpath only at nodes that belong to the destination set. In what follows, we first outline the differences between the MA-VWU and the MA-DAMN problems with the help of an example and then formally state the problem definitions.

Illustrative Example

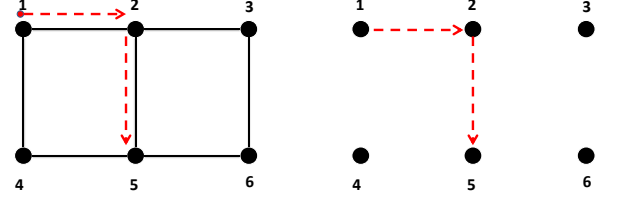
We consider a simple six-node network with bi-directional links. For the sake of simplicity let us assume that the initial network state consists of just a single multicast request $R_1: \{1, (2,5,6), 2\}$. Fig. 1 and 2 demonstrate how the MA-VWU and MA-DAMN models service this request.



(a) Physical topology routing. (b) Lightpaths in the logical overlay.

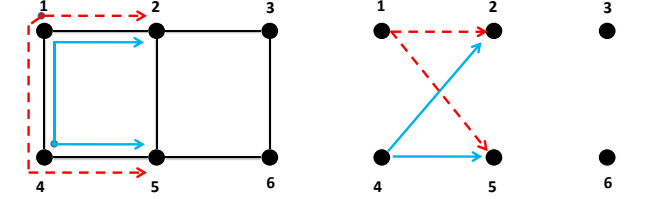
Fig. 1. MA-VWU network state after satisfying request R_1 .

Fig. 1(a) and 1(b) depict the state of the network upon servicing R_1 using the MA-VWU model. More specifically, Fig. 1(a) shows the routing of lightpaths on the physical topology. For simplicity, we select destinations based on their relative proximity (in physical hops) to the source. For R_1 , the selected destinations are nodes 2 and 5. Note that MA-VWU establishes a unique lightpath route to each selected destination. Fig. 1(b) shows the logical view of the lightpaths



(a) Physical topology routing. (b) Lightpaths in the logical overlay.

Fig. 2. MA-DAMN network state after satisfying request R_1 .

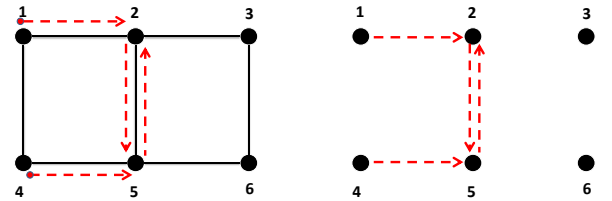


(a) Physical topology routing. (b) Lightpaths in the logical overlay.

Fig. 3. MA-VWU network state after satisfying request R_2 .

reserved in Fig. 1(a). Similarly, Fig. 2 demonstrates the servicing of R_1 according to the MA-DAMN model at both the (a) physical layer, and (b) logical layer. Note that in this case, the multicast request is serviced using multiple hops in the logical (overlay) layer. It is also important to note that the lightpaths are allowed to originate/terminate only at destination nodes of the multicast request. From Fig. 1 and 2, it is easy to verify that both models require only a single wavelength to service R_1 .

Given the current network states for MA-VWU and MA-DAMN, consider the arrival of the second request $R_2: \{4, (2,3,5), 2\}$. Fig. 3 and 4 show how this request can be satisfied using MA-VWU and MA-DAMN respectively. Once again, Fig. 3(a) shows MA-VWU's physical routing, while Fig. 3(b) shows its logical routing. Note that MA-VWU will route from the source node 4 to selected destinations 2 and 5 using independent lightpaths to each destination. Since R_1 already routed to nodes 2 and 5 along wavelength 1 (shown with a broken line), R_2 must be serviced using a second wavelength (shown with a solid line) according to the wavelength conti-



(a) Physical topology routing. (b) Lightpaths in the logical overlay.

Fig. 4. MA-DAMN network state after satisfying request R_2 .

nuity constraint (WCC)⁵ and the wavelength clash constraint⁶. MA-DAMN however, can service R_2 without the need for a second wavelength as shown in Fig. 4. MA-DAMN therefore requires only a single wavelength to service both multicast requests, compared to the required two wavelengths under the MA-VWU approach. However, this decrease in the number of wavelengths comes at the expense of an increase in the average number of logical hops (2 for MA-DAMN, 1 for MA-VWU), which results in a greater end-to-end delay.

For this specific example, MA-DAAN too requires just a single wavelength to service both the requests, however, in some cases MA-DAAN tends to outperform MA-DAMN. Note that we aim to minimize the network wide wavelength count, not the maximum number of wavelengths on any single link. In MA-DAMN and MA-DAAN we service a given multicast request by setting up lightpath routes in the overlay network, wherein we potentially terminate the lightpath at some intermediate node. This incurs an O-E-O conversion at that node, and it is thus fair to allow a free wavelength conversion at that node. For a given lightpath however, we do adhere to the WCC.

The problem statements can now be formally stated follows:

Definitions: Given a network $G = (V, E)$, and a dynamic arrival process for the multicast requests, the **MA-VWU** model must assign, for each request, a lightpath and a wavelength from the source node of the request to K' destination nodes, in such a manner that the number of wavelengths required is minimized while satisfying the WCC. In the **MA-DAMN** model we must assign, for each request, a set of lightpath-routes (possibly multiple-hop) in the overlay layer, to reach K' destination nodes, in such a way that the number of wavelengths required is minimized while satisfying the WCC. The solution must also take into consideration that for any given multicast request, no lightpath terminates/originates at a non-member node (i.e., not a candidate destination node) of the request. The **MA-DAAN** problem is similar to MA-DAMN, with the exception that we now provide the flexibility to terminate/originate a lightpath at/from any non-member of the request set.

III. MANYCAST OVERLAY HEURISTICS

In this section we propose a heuristic for providing multicast capability over *split-incapable* optical WDM networks. The *ManyCast - Shortest Path Overlay (MA-SPO)* heuristic described in Algorithm 1, can be applied to solve both the MA-DAMN and MA-DAAN problems with subtle differences as outlined below.

The input to MA-SPO is a set of predetermined routes, *routeList*, which contains the set of all shortest-path (SP) routes from one multicast request member to every other member in the case of MA-DAMN, or the set of all unicast SP routes for every network node pair in the case of MA-DAAN. The routes in this list are sorted on Line 3 of Algorithm 1, with

Algorithm 1: Multicast Shortest Path Overlay (MA-SPO)

input : Multicast Request: $R = (s_r, D_r, K')$,
 $D_r = \{d_1, d_2, \dots, d_K\}$
: List of appropriate SP routes: *routeList*
output: Multicast overlay tree yielding the fewest number of additional wavelengths to the network

```

1 AltTrees[ $K$ ] = NULL
2 routedDestinations = 0
3 sort(routeList)
4 for each  $d_i \in D_r$  do
5   Tree $k$  = NULL
6   Tree $k$ .add(SP( $s, d_i$ ))
7   update routedDestinations
8   for each route $i$   $\in$  routeList do
9     if routedDestinations  $\geq K'$  then
10      break
11     if (route $i$ .source  $\in$  Tree $k$ ) & (route $i$ .dst  $\notin$ 
12      Tree $k$ ) & (route $i$ .dst  $\in D_s$ ) then
13        Tree $k$ .add(route $i$ )
14        update routedDestinations
15   AltTrees.add(Tree $k$ )
16 return min(AltTrees)

```

the shortest routes (in terms of physical hop count) first. MA-SPO will generate K -alternate logical multicast trees, where $K = |D_r|$. These alternate trees are stored in an array of size K named *AltTrees* in Line 1. Each of these alternate trees is determined within the loop starting at Line 4. For each multicast tree corresponding to a particular candidate destination d_i , the first logical hop is mandated to be the SP from the multicast source s_r to d_i , and this hop is added to the tree at Line 6. It is possible that this first hop to d_i may actually route through one or more other $d_j \in D_r$, and this appropriate number of successfully reached multicast destinations is updated on Line 7. MA-SPO will then inspect each route in the now sorted *routeList* to determine its suitability for inclusion in the multicast tree. In order for a given route to be included in the tree, its source node must already be in the tree, and its destination must not. However, the route's destination must be a member of D_r . The first route in *routeList* which satisfies these criteria will be added to the tree as in Line 12. This continues until K' destinations have been routed to along a single tree, at which point the tree will be added to the collection of alternate trees as in Line 14. Once K -alternate trees (each containing at least K' destination members, if the tree is not empty) have been identified, the least-cost tree is selected as the multicast overlay tree which will satisfy the request in Line 15. The least-cost tree is the one which will result in the smallest number of additional wavelengths required to be added to the current network in order to satisfy the given request. If more than one tree shares this minimum wavelength count, the tie is broken by selecting the tree which incurs the fewest logical hops from s_r to all K' destinations. In the case of MA-DAMN, all routes in *routeList* will originate at a member node and terminate at

⁵A lightpath must occupy the same wavelength across every physical link of the network it traverses.

⁶No two lightpaths may occupy the same wavelength simultaneously.

another member node, thus a signal may be dropped only at a member of the multicast request. In the MA-DAAN case however, any node already included in the tree may be selected as the Steiner point for the next lightpath to be sourced from, thereby relaxing the MA-DAMN constraint that lightpaths may only originate at either the source or destination members, and increasing the routing flexibility of MA-SPO.

In Fig. 5, we present an example to illustrate how the MA-SPO generates K -alternate trees for a given multicast request. Shortest paths between nodes are ordered according to a simple bakery algorithm. For example, if node 1 is to route to node 5, it will route along node 2 rather than node 4 since node 2 has a lower index. Similarly assume that the *routeList* is ordered, such that if a route from s_r to a destination d_i has the same weight as a route from d_j to d_i , the route from s_r is chosen over its competitor. For the multicast request $R = 2:\{4,5,6\}:2$, $K = 3$ alternate trees will be generated by the MA-SPO algorithm, but only $K' = 2$ destinations must be reached in each tree. The first tree requires that its first hop be from node 2 to node 4. The corresponding route is via node 1. The next shortest route is directly from node 2 to node 5. These two paths complete the first tree. The second tree will first route from node 2 to node 5. This now means that the shortest paths to both nodes 4 and 6 are sourced at the closest member, in this case node 5. Relying on the previously mentioned bakery algorithm structure, node 4 is the one selected as the second multicast destination. The first hope of the final tree is mandated to route from node 2 to node 6. The appropriate path is via the intermediate node 3. The second destination in this tree is node 5 which is one hop from both nodes 2 and 6. According to the order of routes in *routeList*, the source for this route should be the source node 2. These three distinct alternate trees are shown in Fig. 5. The current state of the network largely affects which of these trees is chosen for satisfying the multicast request. We select the tree that yields the fewest additional wavelengths to the network. Let us assume that trees 1 and 2 require no new wavelengths, and tree 3 requires 1 new wavelength. Tree 3 is no longer in contention, and the tie is broken by the fact that tree 2 contains 1 logical hop to destination node 5, but two hops to node 4 for a total of 3 logical hops, as opposed to tree 1 which only contains 2 logical hops. For the given request, tree 1 is selected and added to the network.

Complexity: For a network with V nodes, and E links, without considering the complexity of the underlying unicast lightpath reservations for an overlay multicast tree, the complexity of identifying a single tree can be represented by the product of the number of routes traversed in *routeList*, $O(V^2)$, and the maximum number of nodes to traverse in a single tree, $O(V)$. Considering MA-SPO establishes K -alternate trees, the worst-case complexity of Algorithm 1 can be expressed as $O(KV^3)$.

In the following section, we compare the performance of the MA-SPO heuristic to a simple *ManyCast-Shortest Path Unicast* (MA-SPU) heuristic, which provides a sub-optimal solution to the MA-VWU problem by establishing individual unicast

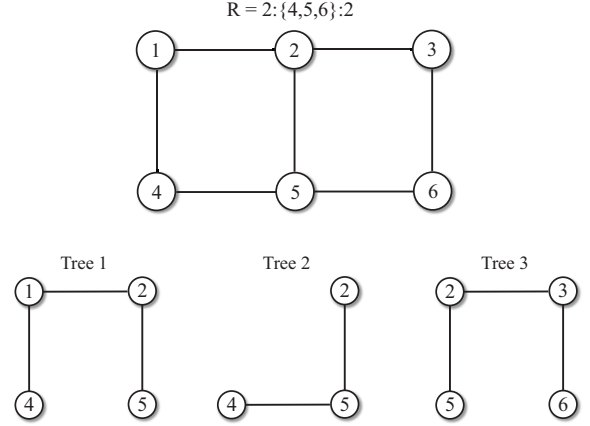


Fig. 5. Illustrative example of MA-SPO for a specific multicast request.

lightpaths from the source node to the *nearest* K' candidate destination nodes. Due to space restrictions, we did not include a full description of the MA-SPU.

IV. SIMULATION RESULTS

We present simulation results of the MA-SPO and MA-SPU heuristics described in the previous section for dynamically arriving multicast connection requests on the National Science Foundation Network (NSFnet) and an augmented Energy Sciences Network (ESnet) shown in Fig. 6.

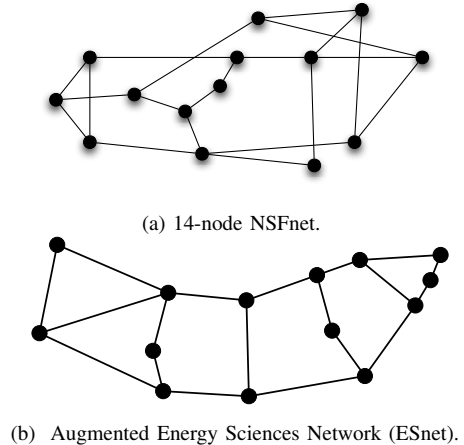


Fig. 6. Networks used for heuristic evaluation.

ManyCast requests arrive according to a Poisson process with average arrival rate λ and exponentially distributed holding times with an average service rate μ . Each request set consists of 10^5 manyCast requests. The results presented in this section represent the average of 30 unique request sets. The source node of each request is uniformly distributed over all nodes in the network. For each manyCast request, the size of the candidate destination set (K) is uniformly distributed across the interval $[3, D_{max}]$ (where D_{max} is a parameter which represents the maximum number of manyCast

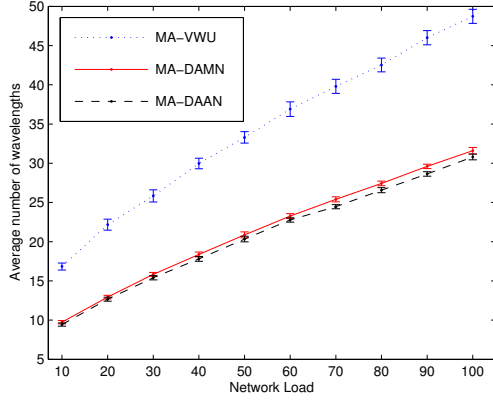


Fig. 7. NSFnet: Average number of wavelengths ($D_{max} = 10$).

destinations). Therefore, for a particular multicast request r , $|D_r| = K$. For each request, the destination nodes are also uniformly distributed across the network. The minimum number of destination nodes (K') to reach is set to $\lceil \frac{|D_r|}{2} \rceil$. We simulate each heuristic with different values for D_{max} , and record the average number of wavelengths. We also plot the 95% confidence intervals for each heuristic. Note that each fiber of the network is assumed to have sufficient number wavelengths to eliminate blocking. The network load in Erlangs is calculated as the ratio of the average arrival rate to the average service rate (λ/μ). Note that we will hereafter refer to the MA-SPO implementations of MA-DAMN and MA-DAAN as simply MA-DAMN and MA-DAAN. Similarly, we will refer to the MA-SPU heuristic as MA-VWU.

In Fig. 7 and Fig. 8 we show the average number of wavelengths required by the three heuristics for the NSFnet and ESnet for $D_{max}=10$. For the NSFnet it is observed that the MA-DAMN achieves approximately 38% improvement (in terms of the number of wavelengths) over MA-VWU. Similarly in Fig. 8, for the ESnet it is observed that MA-DAMN achieves approximately 47% improvement over MA-VWU.

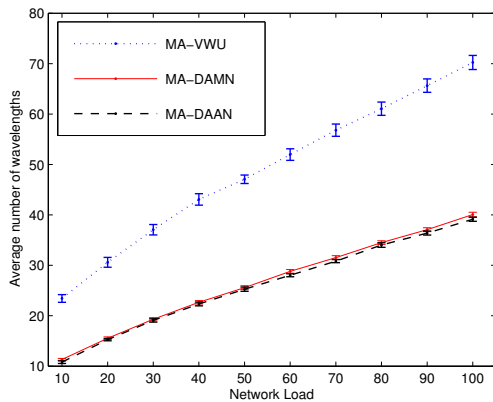


Fig. 8. ESnet: Average number of wavelengths ($D_{max} = 10$).

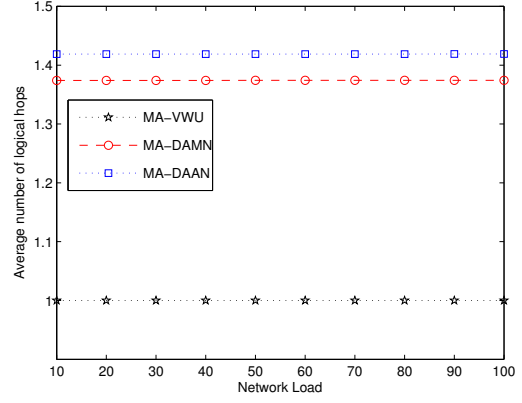


Fig. 9. ESnet: Comparison of logical hops ($D_{max} = 10$).

For both the networks it is seen that MA-DAAN achieves marginal performance improvement over MA-DAMN. This is at the expense of a slightly higher average number of logical hops, as shown in Fig. 9 and Fig. 10 respectively. Note that the MA-VWU heuristic establishes end-to-end to lightpaths from the source node of a request to each selected candidate destination node. Hence the average number of logical hops is always 1.

In Fig. 11 we plot the average runtimes (for successfully servicing all 10^5 multicast requests) for the three heuristics on the ESnet. It can be verified that the run times for MA-VWU are in the order of few tens of seconds. The MA-SPO heuristics not only create logical lightpath trees in the overlay network, but generate K -alternate trees for every request, and then select the best tree for use in establishing lightpaths on the underlying layer. This increases the computational complexity and it can be observed that the MA-DAMN and MA-DAAN implementations of MA-SPO take close to two hundred seconds to complete. Due to space restrictions, we do not show the execution times for the NSFnet, but the results follow a similar trend to those of the ESnet.

In Tables I – IV, we show the results obtained for the

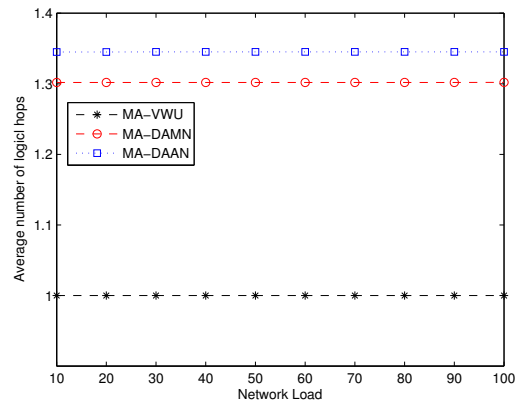


Fig. 10. NSFnet: Comparison of logical hops ($D_{max} = 10$).

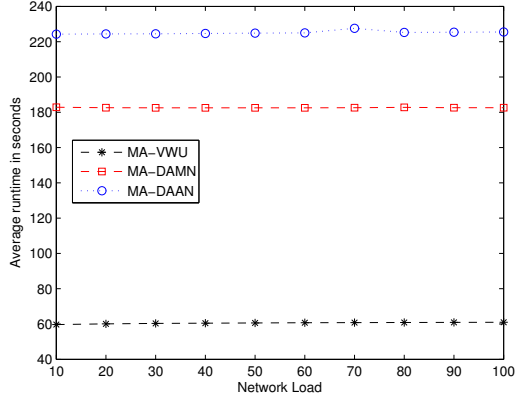


Fig. 11. ESnet: Comparison of run times ($D_{max} = 10$).

TABLE I
NSFNET: AVERAGE NUMBER OF WAVELENGTHS ($D_{max} = 6$).

Load	MA-VWU	MA-DAMN	MA-DAAN	%MA-DAMN	%MA-DAAN
10	13.13	9.33	8.57	28.93	34.77
20	16.80	12.20	11.33	27.38	32.54
30	20.07	14.50	13.73	27.74	31.56
40	22.87	16.80	15.63	26.53	31.63
50	25.27	19.10	17.70	24.41	29.95
60	28.27	21.10	19.43	25.35	31.25
70	30.67	22.97	21.37	25.11	30.33
80	33.20	24.73	23.07	25.50	30.52
90	35.90	26.77	24.90	25.44	30.64
100	37.97	28.67	26.53	24.50	30.11

NSFnet and ESnet for $D_{max} = 6, 8$ respectively. We show the percentage improvement (in terms of wavelength usage) of MA-DAMN (%MA-DAMN) and MA-DAAN (%MA-DAAN) as compared to MA-VWU.

For both networks, it can be observed that MA-DAMN clearly out-performs MA-VWU, and achieves a 30 – 45% improvement. The MA-DAAN achieves a further 2 – 6% improvement over the MA-DAMN. As the number of manycast destinations increase, the performance of both MA-SPO implementations improve relative to MA-VWU, while the improvement of MA-DAAN over MA-DAMN remains relatively consistent across network loads and request set sizes.

TABLE II
ESNET: AVERAGE NUMBER OF WAVELENGTHS ($D_{max} = 6$).

Load	MA-VWU	MA-DAMN	MA-DAAN	%MA-DAMN	%MA-DAAN
10	17.10	10.80	10.37	36.84	39.38
20	22.33	14.77	14.23	33.88	36.27
30	27.27	18.00	17.63	33.99	35.33
40	31.30	21.40	20.90	31.63	33.23
50	35.50	24.10	23.53	32.11	33.71
60	39.53	27.07	26.23	31.53	33.64
70	43.07	29.83	29.10	30.73	32.43
80	46.00	32.53	31.57	29.28	31.38
90	49.73	34.90	34.10	29.83	31.43
100	53.30	37.47	36.50	29.71	31.52

TABLE III
NSFNET: AVERAGE NUMBER OF WAVELENGTHS ($D_{max} = 8$).

Load	MA-VWU	MA-DAMN	MA-DAAN	%MA-DAMN	%MA-DAAN
10	15.48	9.57	8.97	38.21	42.09
20	19.93	12.63	12.27	36.61	38.45
30	23.41	15.27	14.73	34.80	37.07
40	27.55	17.60	16.93	36.12	38.54
50	30.66	19.57	19.10	36.17	37.69
60	33.59	21.90	20.87	34.79	37.87
70	36.72	23.63	22.80	35.65	37.92
80	39.03	25.77	24.67	33.99	36.81
90	42.14	27.50	26.53	34.74	37.03
100	44.66	29.70	28.13	33.49	37.00

TABLE IV
ESNET: AVERAGE NUMBER OF WAVELENGTHS ($D_{max} = 8$).

Load	MA-VWU	MA-DAMN	MA-DAAN	%MA-DAMN	%MA-DAAN
10	20.10	11.00	10.70	45.28	46.78
20	27.48	14.93	14.37	45.66	47.72
30	32.62	18.50	17.97	43.29	44.92
40	37.28	21.47	21.07	42.41	43.48
50	42.21	24.40	24.23	42.19	42.58
60	46.00	27.53	26.80	40.14	41.74
70	50.76	30.17	29.50	40.57	41.88
80	55.07	33.27	32.33	39.59	41.29
90	58.86	35.77	34.80	39.24	40.88
100	63.48	38.47	37.30	39.41	41.24

V. CONCLUSION

The steady emergence of bandwidth-intensive applications effects the need for manycast communication essential in optical WDM networks. Split-incapable networks do not support all-optical manycast splitting, thus creating the problem of establishing the same functionality as a virtual overlay to the unicast-only optical layer. In this paper, we have presented two heuristic solutions to the manycast overlay problem, MA-DAMN and MA-DAAN. Through extensive simulations, we have demonstrated their potential to substantially reduce wavelength utilization (33 – 45%) across large-scale networks over naïve single-hop unicast routing (MA-VWU) to multiple destinations, for realistic, dynamic traffic patterns.

REFERENCES

- [1] R. Jain, "Internet 3.0: Ten problems with current Internet architecture and solutions for the next generation," in *Proc., IEEE MILCOMM*, Oct. 2006.
- [2] M. D. Leenheer, F. Farahmand, K. Lu, T. Zhang, P. Thysebaert, B. Volckaert, F. D. Turck, B. Dhoedt, P. Demeester, and J. P. Jue, "Anycast algorithms supporting optical burst switched grid networks," in *Proc. IEEE International Conference on Networking and Service, (ICNS-'06)*, Silicon Valley, CA, USA, Jul. 2006, pp. 63–69.
- [3] X. H. Q. She, V. M. Vokkarane, and J. P. Jue, "Manycasting over optical burst-switched networks," in *Proceedings, IEEE ICC*, Jun. 2007.
- [4] L. H. Sahasrabudde and B. Mukherjee, "Light trees: optical multicasting for improved performance in wavelength routed networks," vol. 37, no. 2, pp. 67–73, Feb 1999.
- [5] "Advanced networking for distributed petascale science: R&D challenges and opportunities," 2008. [Online]. Available: <http://www.er.doe.gov/ascr/ProgramDocuments/ProgDocs.html>
- [6] H. Takahashi and A. Matsuyama, "An approximate solution for the steiner problem in graphs," *Math. Japonica*, 1980.
- [7] A. Gadkar, J. Plante, and V. M. Vokkarane, "Manycasting: Energy-efficient multicasting in wdm optical unicast networks," in *Proc. Globecom*, Dec. 2011.